

Trainings ▾

General Information

This procedure outlines the proper use of oil analysis techniques to collect an oil sample, to set oil analysis flag limits, and to identify and take appropriate corrective action based on oil analysis results.

Used oil analysis can be employed to monitor engine oil contaminant levels, which can provide evidence of system weaknesses or malfunctions, such as faulty air filtration, coolant leaks, fuel dilution, and wear metals that can suggest engine damage or unusual wear. Oil analysis can be used to proactively identify and address these issues.

Standard oil analysis has limitations, which include the inability to detect large particles and the inability to identify rapidly developing failures. Because of the method used to detect wear metals for standard oil analysis, particles greater than 5 microns (μm) in size can **not** be detected. However, if large particles are suspected, oil analysis labs can perform a ferrography analysis to quantify individual particles in the sample. Properties like magnetism, heat treatment, shape, and size can be identified to help determine the source of the particles and the associated wear mechanism.

The second limitation to oil analysis is the inability to detect rapidly developing failures. In order to identify a potential problem, a failure would have had to be in progress at the time the last oil sample was taken. Some failure modes progress so quickly that they will **not** be identified in oil analysis.

The guidelines outlined in this procedure are **not** to be used to establish oil drain intervals. The use of these guidelines for the purpose of extending oil drain intervals may result in the continued use of oil that no longer provides the intended protection. Oil analysis trends can be charted over a period of time and can provide significant information concerning oil change intervals. For more information on this topic, refer to Fluids for Cummins® Products Service Manual, Bulletin 5411406. See Section 4, or the Operation and Maintenance manual for the specific engine model for other oil related information.

Test Parameters

Table 1 lists those wear metals, oil contaminants, and oil properties that provide the most reliable indicators to detect a potential problem. This list provides both a minimum set of items that should be checked during oil analysis and a list of recommended items to monitor for

extended testing. Extended testing is recommended when a problem with the oil condition is suspected, either from the results of the minimum testing or some other indicator in engine operation.

Table 1: Elements to Monitor for High Horsepower Engine Oil Analysis

Oil Property	Abbreviation	Minimum Testing	Extended Testing
Aluminum (ASTM D5185)	Al	X	X
Copper (ASTM D5185)	Cu	X	X
Fuel dilution (ASTM D7593)	Fuel	X	X
Iron (ASTM D5185)	Fe	X	X
Lead (ASTM D5185)	Pb	X	X
Potassium (ASTM D5185)	K	X	X
Silicon (ASTM D5185)	Si	X	X
Sodium (ASTM D5185)	Na	X	X
Viscosity at 100°C [212°F] (ASTM D445)	Visc	X	X
Total base number (ASTM D4739) ¹	TBN		X
Total acid number (ASTM D664)	TAN		X
Soot (ASTM E1131 ²)	Soot		X
Nitration (ASTM E2412)	Nit		X
Oxidation (ASTM E2412 ³ or D7414 ³)	Ox		X

¹ASTM method D2896 is **not** be used. TBN number can be falsely exaggerated for used engine oil.

²Labs may report a value from E2412 (FTIR) calibrated to E1131.

³Fourier transform infrared spectroscopy (FTIR) data can use either ASTM method but should report whether trending or subtraction.

ISO 17025 is an international standard for testing and calibration laboratories that covers both the methodology and testing equipment of a lab. This standard uses an accreditation body to test if the lab consistently produces valid results. Choosing an oil analysis lab that meets the ISO 17025 certification for the particular oil analysis tests listed above is recommended.

Collecting a Sample

Oil sample collection intervals **must** be set in such a manner that trend comparisons can be made. As oil circulates in an engine during normal operation, wear metals and contaminants accumulate at a steady rate. The amount of contaminants in the oil at the time of sampling generally depends on the length of time since the last oil change. In order to be able to detect changes in the wear pattern of an engine, the sample **must** be collected in consistent hour intervals. Background data is required if oil analysis is to be used correctly. Such data **must** include:

Information to provide with each sample:

- Engine model/serial number
- Miles/hours of oil use
- Miles/hours on engine since new or rebuild
- Oil used (brand name, performance category and viscosity grade)
- Date sample was collected
- Engine application
- Amount of new oil added since previous oil change.

Information to investigate potential issues:

- Any recent engine maintenance
- Analysis of new (unused) oil.

It is important to conduct oil analysis on new (unused) oil to establish a baseline. New (unused) oil analysis samples should be taken each time the oil type or oil supplier is changed or, at a minimum, twice a year. Samples should be taken from the bulk supply tanks to determine the makeup of the oil and also to confirm that no contaminants are being introduced by the storage system.

The sample to be used for analysis **must** be representative of the oil in the engine. Use the following guidelines when collecting oil samples:

- Bring the engine to operating temperature prior to sampling. This will make sure representative contaminant levels are in the sampled increment.
- Successive samples **must** be taken in the same manner and from the same location.
- Take the oil sample before adding any new oil to the engine.
- **Always** collect oil in a clean, dry container.
- Collect a minimum of 118 ml [4 fl-oz] of oil.

Two methods can be used to collect oil samples:

The recommended method for collecting an oil sample is to take the sample from a pressurized port while the engine is idling and warm. This method will make sure the oil sample is **not** stagnant and represents the actual homogeneous oil mixture that is flowing through the engine.

- Clean the outside of the valve by wiping with a clean, dry rag.

- Idle the engine and bring to warm temperature.
- Purge the sample fitting by allowing a small amount of oil to flow through the valve.
- Collect the sample into a clean, dry bottle from the oil stream being pumped by the idling engine.

Note : A recommended location for sampling is a port that allows oil to be collected before the oil is filtered. Reference the Operation and Maintenance Manual for component locations.

In cases where the engine is **not** operational, an alternative method may be used. This method uses a vacuum to draw oil out of the sump. The sample should be obtained as soon after stopping the engine as possible so the oil is still warm and stratification has **not** occurred.

1. Use the dipstick to determine the oil level in the pan.
2. Hold a new, clean piece of tubing against the dipstick and mark, on the tube, the location where the dipstick seats.
3. Cut the tubing so it reaches 25 to 50 mm [1 to 2 in] below the oil level in the pan.
4. Insert the tubing into the dipstick tube so that the mark previously made on the tubing is aligned with the top of the opening for the dipstick tube.
5. Use a hand operated vacuum pump to pump the sample into a clean dry bottle.

Do **not** allow the tube to draw oil from the bottom of the oil pan, because excess debris will be included that could bias the sampling results. Do **not** reuse the sampling tube.

Flag Limits

Understanding the oil system is crucial to identifying when an oil sample is indicating a problem. Large oil system capacity, use of Centinel™ oil replenishment systems, and use of eliminator or centrifuge filtration systems will result in overall lower contamination levels, compared to an engine that has a standard sump capacity, no replenishment system, and standard oil filters.

System capacity can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick and the volume of any remote oil tanks on the machine in which oil is continuously circulated. Oil sump capacities are listed in the operation and maintenance manuals for all Cummins® engines. If the machine is equipped with an oil reserve system with a reservoir remote from the engine oil sump, the reservoir volume **must** be added to the engine sump volume to determine the total system capacity. This is **only** true for remote tanks in which the oil is continuously circulated. The Centinel™ make-up tank volume does **not** add to system capacity, since the engine oil is **not** continuously circulated through this tank.

The addition of new oil will also decrease the overall level of contaminants. In all cases, oil consumption should be documented by the amount of oil added prior to each service interval. Trending oil consumption is important, as any rising trend or step change in oil consumption can indicate possible power cylinder wear, system leaks, or other system malfunction.

One way to compensate for the differences in oil systems between units is to look at the normal wear rates for each engine individually. This method establishes a baseline specific to that particular engine. Provided the oil usage remains fairly constant and other systems are functioning consistently, the trends in wear metals should remain consistent.

For example, setting up statistical flag limits based on a large group will normalize the data. If an engine historically had very low rates of wear, but suddenly jumps to a higher wear rate that is consistent with the average wear rate for the group of engines being analyzed, no flag would be triggered. In comparison, looking at historical readings on a particular unit may cause action to be taken at a lower contaminant level.

Cummins Inc. recommends working with an oil analysis lab to establish warning and critical flag limits using ASTM method D7720-11. If samples are taken periodically throughout the oil drain interval, the oil parameter value can be converted into a rate of increase over new oil. For example, parts per million (ppm) per hour.

As a guideline, engines should be grouped in the following ways for statistical limit comparison:

- Engine Model
- Application
- Similar Duty Cycles
- With Centinel™ and without Centinel™
- With Eliminator™ or Centrifuge type filtration and with standard filter configurations
- Common oil system capacities

If all engines are grouped together regardless of oil systems, the engines with Centinel™, Eliminator™/Centrifuge type filtration, and higher oil system capacities will experience a higher degree of engine wear before the flag limit is reached.

Table 2 lists condemnation limits for oil. If these limits are reached, the oil **must** be changed.

Table 2: Condemnation Limits for High Horsepower Engine Oil Analysis			
Oil Property	Abbreviation	Units	Condemnation Limit
Fuel dilution	Fuel	percent (%)	5%
Viscosity change at 100°C [212°F]	Visc	centistokes (cSt)	For xW-40 - MIN: 12.5, MAX: 16.3
			For xW-30 - MIN: 9.3, MAX: 12.5
Total base number	TBN	milligrams potassium hydroxide per gram of sample (mg KOH/g)	2.5 number minimum or equal to total acid number (TAN)
Total acid number	TAN	milligrams potassium hydroxide per gram of sample (mg KOH/g)	2.5 increase over new oil or equal to total base number (TBN)
Soot	Soot	percent (%)	5% ²

Table 2: Condemnation Limits for High Horsepower Engine Oil Analysis				
Oil Property	Abbreviation	Units	Condemnation Limit	
Nitration	Nit	absorbance units per centimeter (abs/cm)	25	
Oxidation	Ox	absorbance units per centimeter (abs/cm)	15 ¹	
¹ Increase over new (unused) oil sample.				
² Oils meeting API CH-4 (CES 20071) represent a minimum quality standard for Cummins® high horsepower engines. Oils meeting older API requirements (C F-4, CG-4) are not formulated to tolerate greater than 1.5 percent soot.				
Table 3 is to be used only as a reference if flag limits are not able to be established through statistical analysis.				
Table 3: Flag Guidelines for High Horsepower Engine Oil Analysis				
Oil Property	Abbreviation	Units	Caution Flag Limit	Critical Flag Limit
Aluminum	Al	parts per million (ppm)	15	30
Copper	Cu	parts per million (ppm)	60	180
Iron	Fe	parts per million (ppm)	50	130
Lead	Pb	parts per million (ppm)	20	50
Potassium	K	parts per million (ppm)	20 ¹	110 ¹
Silicon	Si	parts per million (ppm)	15 ¹	110 ¹
Sodium	Na	parts per million (ppm)	20 ¹	110 ¹
¹ Increase over new (unused) oil sample.				

Interpreting Results The troubleshooting trees listed below are intended to serve as a guide for investigating instances of high wear metals or oil contaminants. Some of the trees are set up to direct investigation based on the severity level of the flag. The caution severity level indicates that an abnormal wear condition exists, and the investigation may require minor maintenance to be performed.
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The critical severity level indicates that major wear or contamination exists that requires immediate attention, and the investigation may require partial engine disassembly.

These troubleshooting trees are organized so that a problem can be located and corrected by doing the easiest and most logical things first. Complete all steps in the sequence shown from top to bottom.

Some of the steps in the troubleshooting trees require information on oil drain intervals, recent repairs, coolant makeup or oil makeup. This information is used to verify that the engine is experiencing a true incident of high wear or contamination and **not** an abnormal oil condition due to some other factor.

Many of the troubleshooting trees use combinations of wear metals or contamination indicators to increase confidence in determining where the problem is originating. See the list below to become familiar with sources of wear metals or contaminants.

Sources of Wear Metals or Contaminants	
(Al) Aluminum	Pistons (some applications), crankshaft and camshaft thrust bearings, camshaft bushings, dirt.
(Cu) Copper	Connecting rod and main bearings, turbocharger bearings, connecting rod bushings, camshaft bushings, idler gear and accessory drive bushings and thrust washers, cam follower pins, rocker lever bushings, lube pump bushings and thrust washers, oil coolers.
Fuel	Unburned fuel in lubricating oil, injectors or fuel pump can be source of contamination. Engine operation, such as frequent starts, excessive idling, and cold weather operation can also lead to fuel dilution.
(Fe) Iron	Cylinder liner, pistons (some applications), gears, crankshaft, camshaft and camshaft follower rollers and shafts.
(Pb) Lead	Rod and main bearings, turbocharger bearings, camshaft bushings, idler gear and accessory drive bushings and thrust washers, lube pump bushings and thrust washers.
(K) Potassium	Element contained in some coolant additive packages, which indicates coolant contamination of oil.
(Si) Silicon	Sand, dirt, silicon sealants and gasket materials, oil additives, cooling system inhibitor.

Sources of Wear Metals or Contaminants	
(Na) Sodium	Element contained in some coolant additive packages which indicates coolant contamination of oil, salt from detergent or road, oil additive.
Viscosity	The property of a fluid measured at a certain temperature that resists the force to cause the fluid to flow.
(TAN) Total Acid Number	A measure of oil's acidity. New oil has a certain level, or starting TAN, due to the type and amount of additives.
(TBN) Total Base Number	A measure of the oil's ability to neutralize acid. New oil has a certain level in the oil formula that is depleted over time due to contact with acid generated by blowby gases.
Soot	Combustion by-products in oil.
Oxidation	A generic measure of the breakdown of a lubricant due to age and exposure to high temperature. High oxidation numbers indicate oil is beyond its useful life.
Nitration	A generic indicator of excessive "blowby" and also indicates the presence of nitric acid formed by nitric oxide, which is a combustion by-product when the engine operates too lean. Nitric acid can attack metal surfaces in the engine and result in corrosive wear. Nitric acid can accelerate oil oxidation.
Oil Analysis Flags for Common Failure Modes: Additional information can be obtained when looking at combinations of wear metals and/or contaminants. Listed below are some engine problems that may result in oil analysis flags and an explanation of what combination of elevated wear metals and/or contaminants to look for if suspected.	
Engine Problems That Result in Oil Analysis Flags	
Fuel Dilution	The best indicators of fuel dilution are an increased percentage of fuel in the oil and a decrease in viscosity. In cases of high fuel dilution, the oil may smell like fuel and the engine oil level may increase.

Engine Problems That Result in Oil Analysis Flags

Coolant Contamination

The best method to detect coolant contamination of oil is to look for traces of elements, such as sodium and potassium, which are contained in the coolant additive package. These elements can be detected at very low levels, so they can serve as an early warning. Sodium is occasionally found in lubricant additive packages; therefore, it is important to compare the level of sodium with that measured in the baseline (unused) oil sample and to be aware of the coolant additive package formulation. A water content test is **not** a reliable indicator of a coolant leak because oil temperatures are generally higher than the boiling point of water, which allows the water to boil out during normal engine operation, leaving **only** the elements that make up the coolant additive package behind. Glycol tests can also be inaccurate. Small amounts of glycol may **not** be measurable, and high oil temperatures can break down glycol to a state where it will **not** be detected. In addition to identifying the source of coolant contamination, it is important to determine if progressive damage has occurred. Coolant contamination of the lubricating oil can damage bearings and bushings.

Dust Out

Dirt is composed primarily of silica and alumina, so elevated levels of silicon and aluminum in a 2:1 to 10:1 Silicon:Aluminum ratio indicate dirt intrusion. In addition to identifying the source of dirt entry, it is important to determine if progressive damage has occurred. A dust out can cause piston ring/cylinder liner wear, which can lead to lubricating oil contamination or a breakdown of the lubricating oil properties and bearing/bushing failures.

Engine Problems That Result in Oil Analysis Flags		
Piston Ring/Cylinder Liner Wear	Increased levels of iron indicate cylinder liner wear. Additionally, wear of the rings and/or liner may allow blowby gases to pass by the rings and contaminate the oil. Elevated levels of soot, oxidation, and nitration along with an increase in viscosity may be observed. Contamination of the oil will also increase total acid number and decrease total base number. In addition to identifying the source of coolant contamination, it is important to determine if progressive damage has occurred. Contaminated lubricating oil can damage bearings and bushings.	
Bearing/Bushing Failure	Bearings are made up primarily of lead and copper. An increased level of both of these metals at the same time indicates failure of a bearing and/or bushing. An increase of lead with no increase in copper typically corresponds to a degradation of oil condition that results in attack of the top layer of the bearing. Some other indications may include increases in oxidation, nitration, TAN and/or a decrease in TBN.	
Note : In order to be alerted to a problem, a failure would have had to be in progress at the time the last oil sample was taken. Some failure modes progress so quickly that they will not be identified in oil analysis. Note : When oil is changed on an engine with a remote oil tank in which oil is continuously circulated (not a Centinel™ make up tank), the oil in the remote tank must be changed, in addition to the oil in the engine sump. If the oil in the remote tank is not changed, it will act as a source of contamination for the next oil sample. Troubleshooting and repair procedures can be found in the appropriate set of manuals on Quickservice™ Online for the engine being investigated. Cummins Inc. does not pay for oil analysis work and will not reimburse for the testing required for an oil analysis program.		
Aluminum parts per million (PPM) level exceeds caution or critical aluminum flag limit based on oil lab results or Cummins Inc. limits		
Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, wear metals may be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.

Aluminum parts per million (PPM) level exceeds caution or critical aluminum flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing an elevated level of wear metals.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.
Engine has aluminum pistons and has had at least one power cylinder (piston) replaced within the last 1000 hours	Repair to power cylinder resulting in break-in wear.	Resample at regular interval. Aluminum should return to normal level within 1,000 hours of operation.
Engine has aluminum pistons and is new or recently rebuilt with less than 1000 hours or operation	Power cylinder break-in results in elevated wear metal levels.	Resample at regular interval. Aluminum should return to normal level within 1,000 hours of operation.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Aluminum piston skirt scuffing	Aluminum piston skirt scuffing.	Check ECM image for blowby faults and increased oil consumption or excessive engine noise. Follow the respective troubleshooting trees.

Aluminum parts per million (PPM) level exceeds caution or critical aluminum flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Dirt intrusion	Dirt is made up of Silica and Alumina. Silicon and Aluminum in 2:1 to 10:1 Silicon:Aluminum ratio indicates dirt.	Check whether silicon is also at elevated level. If so, check if dipstick tube and/or oil fill cap is missing, loose, or has a missing seal. Inspect the intake system for holes and sources of leaks. Check for collapsed air filters and proper sealing around air cleaners. Repair as necessary. Check for increased oil consumption and high blowby. Reference the Lubricating Oil Consumption Excessive or Crankcase Gases (Blowby) Excessive Troubleshooting Tree. Check whether iron, copper and/or lead are also at an elevated level. If so, reference the respective troubleshooting tree.
Increased wear rate of thrust bearings	Increased wear rate of thrust bearings.	Check if crankshaft end clearance is within specification. If not , inspect or repair thrust bearings.

Copper parts per million (PPM) level exceeds copper caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, wear metals can be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing an elevated level of wear metals.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.

Copper parts per million (PPM) level exceeds copper caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Engine has had at least one oil cooler replaced within the last 1000 hours	Oil cooler passivation.	Resample at regular interval. Copper should return to normal level within 1,000 hours.
Engine is new or recently rebuilt with less than 1000 hours of operation	Oil cooler passivation.	Resample at regular interval. Copper should return to normal level within 1,000 hours .
Oil formulation has recently been changed	Oil cooler passivation.	Resample at regular interval. Copper should return to normal level within 1,000 hours.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil	Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	Check if lead is also at an elevated level. If so, cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings. If no debris, change oil, reassess oil drain interval, resample at regular interval.
Corrosive attack of engine bearing material due to coolant contamination	Oil acidity is attacking bearing material due to coolant contamination.	Check if lead is also at an elevated level. If so, cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings. Check whether sodium and/or potassium are also at elevated levels. If so, reference sodium and/or potassium troubleshooting trees.

Copper parts per million (PPM) level exceeds copper caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Wear of bearing material due to reduced oil film thickness	High temperature, low oil pressure, or low viscosity compromise oil film thickness.	Check whether lead is also at an elevated level. If so, check for oil temperature or pressure faults and cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings.

Copper parts per million (PPM) level exceeds critical copper flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Possible lab error, sample contamination, or other reasons causing a single high reading	Possible lab testing error, recording error, sample contamination, or other reasons for a single high reading.	If a sudden high reading, perform the steps in the copper exceeds caution flag limit tree. If lead also has a sudden high reading, cut oil filters open or open centrifuge and check for debris. If no debris, resample at half interval (including TBN measurement) and request ferrography analysis on oil sample that produced elevated copper level.
Engine has had at least one oil cooler replaced within the last 1000 hours	Oil cooler passivation.	Resample at regular interval. Copper should return to normal level within 1,000 hours of operation.
Engine is new or recently rebuilt with less than 1000 hours of operation	Oil cooler passivation.	Resample at regular interval. Copper should return to normal level within 1,000 hours of operation
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.

Copper parts per million (PPM) level exceeds critical copper flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	If Total Base Number (TBN) level is low, evaluate oil drain interval and oil type. If lead is also at an elevated level, cut filters open or open centrifuge and check for debris. If no debris found, request ferrography analysis on oil sample that produced elevated copper level. If debris in filter/centrifuge or ferrography analysis indicates bearing debris, inspect bearings.
Corrosive attack of bearing material due to coolant contamination	Oil acidity is attacking bearing material due to coolant contamination.	Check whether sodium and/or potassium are also at critical levels. If so, reference Coolant Loss - Internal troubleshooting tree. Check whether lead is also at elevated level, if so, cut filters open or open centrifuge and check for debris. If no debris found, request ferrography analysis on oil sample that produced elevated copper level. If debris in filter/centrifuge or ferrographic analysis indicates bearing debris, inspect bearings.

Copper parts per million (PPM) level exceeds critical copper flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Wear of bearing material due to reduced oil film thickness	High temperature, low oil pressure, or low viscosity compromise oil film thickness.	Check for oil temperature or pressure faults. Check whether viscosity is low. If so, reference low viscosity troubleshooting tree. Check whether lead is also at elevated level, if so, cut filters open or open centrifuge and check for debris. If no debris found, request ferrography analysis on oil sample that produced elevated copper level. If debris in filter/centrifuge or ferrographic analysis indicates bearing debris, inspect bearings.
Wear of bushings not containing lead	Connecting rod bushing, gear bushing, cam follower bushings.	Check for oil temperature or pressure faults. Check whether viscosity is low. If so, reference low viscosity troubleshooting tree. Cut filters open or open centrifuge and check for debris. If no debris found in filters/centrifuge, request ferrography analysis on oil sample that produced elevated copper level. If debris in filter/centrifuge or ferrography analysis indicates bushing debris, inspect bushings.

Fuel dilution exceeds flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Frequent Starts	Fuel does not burn completely at low combustion chamber temperatures and can seep past the piston rings and into the crankcase.	Change oil in engine sump and remote oil tank.

Fuel dilution exceeds flag limit based on oil lab results or Cummins Inc. limits		
Cause	Explanation	Correction
Excessive Idling	Fuel does not burn completely at low combustion chamber temperatures and can seep past the piston rings and into the crankcase.	Change oil in engine sump and remote oil tank. If possible, shut off the engine rather than idling or raise the idle speed.
Cold Weather Operation	Fuel does not burn completely at low combustion chamber temperatures and can seep past the piston rings and into the crankcase.	Change oil in engine sump and remote oil tank.
Frequent starts, excessive idling, or cold weather operation	Fuel does not burn completely at low combustion chamber temperatures and can seep past the piston rings and into the crankcase.	Change oil in engine sump and remote oil tank. If possible, shut off the engine rather than idling or raise the idle speed.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Engine has had at least one injector removed within the last 1000 hours	Injector o-rings cut or blown, injector hold down mis-torqued, injector seal mis-installed.	Dye fuel and inspect area around injector with black light. Repair as necessary.
Oil contaminated by fuel due to fuel pump failure or injector malfunction	Excessive fuel getting in lube oil reduced viscosity of lube oil. Oil will have fuel smell only at high fuel dilution levels.	Confirm that viscosity has also dropped below limit. If so, follow Fuel in the Lubricating Oil Troubleshooting Tree.
Iron parts per million (PPM) level exceeds iron caution flag limit based on oil lab results or Cummins Inc. limits		
Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, wear metals may be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.

Iron parts per million (PPM) level exceeds iron caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing an elevated level of wear metals.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.
Engine has had at least one power cylinder (rings and/or liner) replaced within the last 1000 hours	Repair to power cylinder resulting in break-in wear.	Resample at regular interval. Iron should return to normal level within 1,000 hours of operation.
Engine is new or recently rebuilt with less than 1000 hours of operation	Power cylinder break-in wear.	Resample at regular interval. Iron should return to normal level within 1,000 hours of operation.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Wear of interface between steel and bushing/bearing material	High iron and copper and/or lead indicates bearing damage.	Check whether lead and copper are also at elevated levels. If so, reference copper and/or lead troubleshooting trees.
Increased wear rate of rings and/or liners due to fuel washing lubrication from liners	Injector failed, causing cylinder wash down with fuel, which caused liner wear, resulting in higher iron.	Check for evidence of low viscosity or fuel dilution. If so, reference fuel dilution or low viscosity troubleshooting tree.
Increased wear rate of rings and/or liners due to dusting or worn rings	High cylinder wear due to dusting, broken ring, or worn rings.	Check for increased oil consumption. Reference Lubricating Oil Consumption Excessive Troubleshooting Tree.
Increased wear rate of rings and/or liners due to overfueling	High cylinder temperatures caused by excessive fuel or restricted air flow can accelerate cylinder wear.	Check for smoke or low power. If engine smokes or does not pull power, follow Engine Performance Troubleshooting Trees.

Iron parts per million (PPM) level exceeds iron caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Iron to iron wear surfaces in engine	Camshaft, roller, or gear failure generating debris.	Request ferrography analysis on oil sample that produced elevated iron level. Cut oil filters open or open centrifuge and check for debris. If debris in filter/centrifuge or ferrography analysis indicates gear, camshaft or roller debris, inspect accordingly. If no debris, change oil, and resample at regular interval.

Iron parts per million (PPM) level exceeds critical iron flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Possible lab error, sample contamination, or other reasons causing a single high reading	Possible lab testing error, recording error, sample contamination, or other reasons for a single high reading.	If a sudden high reading, perform the steps in the iron exceeds caution flag limit tree. If other metals also have a sudden high reading, continue with this troubleshooting tree.
Engine has had at least one power cylinder (rings and/or liner) replaced within the last 1000 hours	Repair to power cylinder resulting in break-in wear.	Resample at regular interval. Iron should return to normal level within 1,000 hours of operation.
Engine is new or recently rebuilt with less than 1000 hours of operation	Power cylinder break-in wear.	Resample at regular interval. Iron should return to normal level within 1,000 hours of operation.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.

Iron parts per million (PPM) level exceeds critical iron flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Wear of interface between steel and bushing/bearing material	High iron and copper and/or lead indicates bearing damage.	Check whether lead and copper are also at elevated levels. If so, reference copper and/or lead troubleshooting trees.
Increased wear rate of rings and/or liners due to fuel washing lubrication from liners	Injector failure causing cylinder wash down with fuel can cause liner wear resulting in higher iron.	Check for evidence of lower viscosity or fuel dilution. If so, see fuel dilution and/or low viscosity troubleshooting tree.
Increased wear rate of rings and/or liners due to over fueling	High cylinder temperatures caused by excessive fuel or restricted air flow can accelerate cylinder wear.	Check for smoke or low power. If engine smokes or does not pull power, follow Engine Performance Troubleshooting Trees.
Increased wear rate of rings and/or liners due to broken or worn rings	High cylinder wear due to broken ring, worn rings causing higher iron.	Check for increased oil consumption. If so, see Lubricating Oil Consumption Excessive troubleshooting tree.
Increased wear rate of rings and/or liners due to dust intrusion	High cylinder wear due to dusting, broken ring, or worn rings.	Verify if silicon is also at an elevated level. If so, check if dipstick tube and/or oil fill cap is missing, loose, or has a missing seal. Inspect the intake system for holes and sources of leaks. Check for collapsed air filters and proper sealing around air cleaners. Repair the intake system if necessary. Check blowby. If high, follow the Crankcase Gases (Blowby) Excessive troubleshooting tree.

Iron parts per million (PPM) level exceeds critical iron flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Iron to iron wear surfaces in engine	Camshaft, roller, or gear failure generating debris.	Request ferrography analysis on oil sample that produced elevated iron level. Cut oil filters open or open centrifuge and check for debris. If debris in filter/centrifuge or ferrography analysis indicates gear, camshaft or roller debris, inspect accordingly. If no debris, change oil, and resample at regular interval.

Lead parts per million (PPM) level exceeds lead caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, wear metals may be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing an elevated level of wear metals.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	Cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings. If no debris, change oil, reassess oil drain interval, resample at regular interval.

Lead parts per million (PPM) level exceeds lead caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Corrosive attack of bearing material due to coolant contamination	Oil acidity is attacking bearing material due to coolant contamination.	Check if copper is also at an elevated level. If so, cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings. Check whether sodium and/or potassium are also at elevated levels. If so, reference sodium and/or potassium troubleshooting trees.
Wear of bearing material due to reduced oil film thickness	High temperature, low oil pressure, or low viscosity compromise oil film thickness.	Check whether copper is also at an elevated level. If so, check for oil temperature or pressure faults and cut oil filters open or open centrifuge and check for debris. If bearing debris in filter/centrifuge, inspect bearings.

Lead parts per million (PPM) level exceeds critical lead flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Possible lab error, sample contamination, or other reasons causing a single high reading	Possible lab testing error, recording error, sample contamination, or other reasons for a single high reading.	If a sudden high reading, perform the steps in the lead exceeds caution flag limit tree. If copper also has a sudden high reading, cut oil filters open or open centrifuge and check for debris. If no debris, resample at half interval (including TBN measurement) and request ferrography analysis on oil sample that produced elevated lead level.

Lead parts per million (PPM) level exceeds critical lead flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	Oil acidity is attacking bearing material due to low Total Base Number (TBN) in oil.	If Total Base Number (TBN) level is low, evaluate oil drain interval and oil type. Cut filters open or open centrifuge and inspect for debris. If no debris found, request ferrography analysis on oil sample that produced elevated lead level. If debris in filter/centrifuge or ferrography analysis indicates bearing debris, inspect bearings.
Corrosive attack of bearing material due to coolant contamination	Oil acidity is attacking bearing material due to coolant contamination.	Check whether sodium and/or potassium are also at critical levels. If so, reference Coolant Loss - Internal troubleshooting tree. Check whether copper is also at elevated level, if so, cut filters open or open centrifuge and check for debris. If no debris found, request ferrography analysis on oil sample that produced elevated lead level. If debris in filter/centrifuge or ferrography analysis indicates bearing debris, inspect bearings.

Lead parts per million (PPM) level exceeds critical lead flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Wear of bearing material due to reduced oil film thickness	High temperature, low oil pressure, or low viscosity compromise oil film thickness.	Check for oil temperature or pressure faults. Check whether viscosity is low. If so, reference low viscosity troubleshooting tree. Check whether copper is also at elevated level, if so, cut filters open or open centrifuge. If no debris found, request ferrography analysis on oil sample that produced elevated lead level. If debris in filter/centrifuge or ferrography analysis indicates bearing debris, inspect bearings.

Potassium parts per million (PPM) level exceeds caution potassium flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil contaminated by coolant	Potassium is commonly used as a cooling system inhibitor.	Verify that coolant used in this engine contains potassium. Check whether sodium and/or silicon are also at elevated levels if there are also contained in coolant. If so, monitor coolant consumption, check for and repair any head gasket and water pump weep hole leaks and resample at half of normal interval.

Potassium parts per million (PPM) level exceeds critical potassium flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Possible lab error, sample contamination, or other reasons causing a single high reading	Possible lab testing error, recording error, sample contamination, or other reasons for a single high reading.	If a sudden high reading, verify that coolant used in this engine contains potassium. Check if there is an elevated reading in silicon or sodium, if these are contained in coolant. Check for elevated levels of lead or copper. If these combinations also have a sudden high reading, or if oil tests positive for glycol, continue with tree. If only potassium is elevated, follow the potassium exceeds caution flag limit tree.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil contaminated by coolant	Potassium is commonly used as a cooling system inhibitor.	Check for and repair any head gasket and water pump weep hole leaks. If no external leaks or problem persists, follow Coolant Loss - Internal Troubleshooting Tree. Check turbocharger axial and radial clearance.

Silicon parts per million (PPM) level exceeds the caution or critical silicon flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, wear metals may be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.

Silicon parts per million (PPM) level exceeds the caution or critical silicon flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing an elevated level of wear metals.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.
Engine has had a repair where silicone gaskets have been replaced or silicone sealant has been applied within the last 1000 hours	Repair to engine resulting in replacement of silicone gasket or silicone sealant that is reacting with oil.	Resample at regular interval. Silicon should return to normal level within 1,000 hours of operation.
Engine is new or recently rebuilt with less than 1000 hours of operation	Repair to engine resulting in replacement of silicone gasket or silicone sealant that is reacting with oil.	Resample at regular interval. Silicon should return to normal level within 1,000 hours of operation.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Dirt intrusion	Dirt is made up of Silica and Alumina. Silicon and Aluminum in 2:1 to 10:1 Silicon:Aluminum ratio indicates dirt.	Check whether aluminum is also at elevated level. If so, check if dipstick tube and/or oil fill cap is missing, loose, or has a missing seal. Inspect the intake system for holes and sources of leaks. Check for collapsed air filters and proper sealing around air cleaners. Repair as necessary. Check for increased oil consumption and high blowby. Reference Lubricating Oil Consumption Excessive or Crankcase Gases (Blowby) Excessive Troubleshooting Tree. Check whether iron, copper and/or lead are also at an elevated level. If so, reference the respective troubleshooting tree.

Silicon parts per million (PPM) level exceeds the caution or critical silicon flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Coolant contamination	Silicon is commonly used as a cooling system inhibitor.	Verify that coolant used in this engine contains silicon. Check whether sodium and/or potassium are also at elevated levels. If so, reference sodium and/or potassium troubleshooting trees.

Sodium Parts per million (PPM) level exceeds sodium caution flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil formulation has recently been changed	Sodium is sometimes contained in the oil additive package.	Compare parts per million (ppm) of sodium with unused oil sample.
Wash water entered engine contaminating oil	Salts (sodium) are contained in detergents.	Determine if engine could have recently been contaminated with high pressure wash water. If so, change oil and resample at normal interval.
Oil contaminated by coolant	Sodium is commonly used as a cooling system inhibitor.	Verify that coolant used in this engine contains sodium. Check whether silicon and/or potassium are also at elevated levels, if these are also contained in coolant. If so, monitor coolant consumption, check for and repair any head gasket and water pump weep hole leaks and resample at half of normal interval.

Sodium Parts per million (PPM) level exceeds critical sodium flag limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Possible lab error, sample contamination, or other reasons causing a single high reading	Possible lab error, sample contamination, or other reasons causing a single high reading.	If a sudden high reading, verify that coolant used in this engine contains sodium. Check if there is an elevated reading in silicon or potassium, if these are contained in coolant. Check for elevated levels of lead or copper. If these combinations also have a sudden high reading, or if oil tests positive for glycol, continue with tree. If only sodium is elevated, follow the sodium exceeds caution flag limit tree.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil contaminated by coolant	Sodium is commonly used as a cooling system inhibitor.	Check for and repair any head gasket and water pump weep hole leaks. If no external leaks or problem persists, follow Coolant Loss - Internal Troubleshooting Tree. Check turbocharger axial and radial clearance.

Viscosity is higher than limit based on oil lab results or Cummins Inc. limits

Cause	Explanation	Correction
Extended oil sample interval	If oil hours at time of sample are greater than oil hours for past samples, viscosity may be elevated.	Check whether hours on oil at time of sample is more than 25 percent longer than previous sample hours. If so, resample at regular interval.

Viscosity is higher than limit based on oil lab results or Cummins Inc. limits		
Cause	Explanation	Correction
Remote oil tank circulation, makeup oil or Centinel™ systems are not functioning properly	Oil is not being replenished or oil capacity is reduced, causing increased viscosity.	Repair remote oil circulation, makeup oil, or Centinel™ system. Resample at regular interval.
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil formulation and/or viscosity grade has recently been changed	Change in oil viscosity due to difference in oil specification.	Confirm viscosity is within specification of oil formulation. Resample at regular interval.
Coolant emulsion with oil	Oil contaminated by coolant.	Check whether sodium and/or potassium are also at elevated levels. If so, reference sodium and/or potassium troubleshooting trees.
Water emulsion with oil	Combustion by-product condensation occurs with extended light load running and cold weather conditions.	Change oil. Increase oil temperature by running engine at higher loads or using other means and reduce operation at light loads.
Oil contaminated by combustion by-products or dust intrusion	Contamination of oil causing higher viscosity.	Check blowby. If high, follow the Crankcase Gases (Blowby) Excessive troubleshooting tree.
Excessive oil temperature	High oil temperature due to localized component failure causes oil oxidation and increases viscosity. Can also see elevation in wear metals.	Check for increase in oxidation of oil. Check for high oil temperature. If high, follow high oil temperature troubleshooting tree. Check for elevated levels of iron, copper, lead, and/or aluminum. If high, follow troubleshooting trees for respective metals.

Viscosity is lower than limit based on oil lab results or Cummins Inc. limits		
Cause	Explanation	Correction
Engine has recently been repaired, causing oil contamination	Previous repair caused oil contamination and oil system was not purged.	Change oil in engine sump and remote oil tank. Check external oil circuitry (i.e. remote filter head/remote oil tank) for contamination. Resample at regular interval.
Oil formulation and/or viscosity grade has recently been changed	Change in oil viscosity due to difference in oil specification.	Resample at regular interval.
Oil contaminated by fuel	Fuel contamination reduces oil viscosity.	Verify that fuel dilution is also flagged on oil sample. If so, follow Fuel in the Lubricating Oil Troubleshooting Tree.